00: The Staging, the Professor, and the Tribunal of Paper 001

AuthorityEffect: None.
Location: Moscow, Sadovaya 302-bis, Apartment 50.
Presiding Tribunal: Messire Woland, Lev Davidovich Landau ("Dau"), Koroviev-Fagotto, Behemoth, Azazello, Abadonna, Hella; with the Russian-American Delegation: Jeremy, Liudmila, Polina, Yaroslava.
Subject of Examination: Paper 001 — *The Symbol Game: Discrete Cochain Retraction, Primal-Dual Obstruction, and Toy Involutions* (PFP I: Exact Graph Repair and the Boundary Between a Proposal and a Certificate).
Companion Documents: - Study Guide: study_guide.md - Examination Paper: quiz.md - Instructor Key: instructor_key.md - Notebook worksheet: paper001_notebook_learning_worksheet.md

The worksheet has six exercises on the notebooks and the Lean sources. It does not replace the thirteen-problem examination.

This is an imagined literary examination. The dialogue and tribunal are fictional, rather than historical quotations or biographical evidence. The mathematical definitions follow the manuscript.

1. The Knock at the Door: The Second Trial

The sleet over Sadovaya Street had thickened into a foul, freezing slush that plastered the bare lime trees along the Garden Ring. Inside Apartment 50, the atmosphere had shifted from the declared two-mark carrier of Paper 000 into the sharp, metallic tension of a computational workshop.

You had practiced the two-mark exercises. Their labels do not establish a verified transition from $S(0)$ to $S(4)$. But now your leather portfolio bulged with printouts from gradient-descent packages, floating-point loss curves, continuous Laplace-Beltrami operators, and PyTorch autograd scripts. On your scratchpad, you had scribbled the fatal, lazy claim:

"A candidate potential with cost 3.0000001 converges to the global minimum."

You knocked on the black oak door.

The door flew back instantly. **Hella** stood before you, holding a brass caliper in one bloodless hand. Her cold green eyes flicked to your satchel:

> *"PyTorch? SciPy `optimize.minimize`? Floating-point epsilon approximations?*
> *Throw those contaminated papers down the stairwell this instant. In*

Paper 001, we calculate in exact rationals $\mathbb{Q}$. If you attempt to smuggle an IEEE 754 float across this threshold, I will sever your tendons with this caliper."

She stepped aside, gesturing with a contemptuous jerk of her chin toward the candlelit salon.

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2. Woland and Jeremy Carroll

In the center of the drawing room sat **Messire Woland**, leaning on his silver poodle-headed cane. His black eye was fixed upon a large sheet of butcher paper spread across the dining table, upon which was drawn a simple directed triangle:

$$a \to b \quad (w_1 = 1, c_1 = 1), \quad b \to c \quad (w_2 = 2, c_2 = 1), \quad c \to a \quad (w_3 = 3, c_3 = 1)$$

Beside Woland stood **Jeremy Carroll**, the author of Paper 001. Jeremy had his sleeves rolled to his elbows, chalk under his fingernails, and a worn mechanical pencil tucked behind his ear:

> *"Look at him, Messire,"* Jeremy said in a dry, calm Michigan drawl—a true Yooper from Sault Ste. Marie who didn't waste syllables. *"He walked all the way from Cambridge or Silicon Valley with twenty thousand dollars worth of GPU cloud credits, and he thinks he can optimize a three-edge graph by running gradient descent.
>
> He doesn't know the difference between a **proposal** and a **certificate**. He thinks that because he found an assignment of node levels that costs three, he has certified that no assignment costs two. He has never seen a dual circulation in his life."*

Woland turned his mismatched eyes upon you. His green eye flared with cold, ironic mirth:

> *"He believes in algorithms, Jeremy. He believes that if an artificial brain sweats through forty iterations of gradient descent, the resulting floating-point tensor constitutes a mathematical theorem.
>
> Step forward, Lev Davidovich! Show our guest what happens when continuous hand-waving meets discrete graph topology."*

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3. Landau Takes the Slate

Lev Davidovich Landau ("Dau") erupted from behind the chalkboard, his railway chalk spinning in a blur between his fingers. He slammed the chalk

against the slate, producing a shriek that rattled the teacups on Liudmila's tray:

*"Quiet! Let us review the crime!

Three clocks on a wall! The first says b is 1 hour ahead of a. The second says c is 1 hour ahead of b. The third says a is 1 hour ahead of c! You walk around the triangle and you are 3 hours in the future! Can this discrepancy be resolved by assigning consistent times s_a, s_b, s_c?

Of course not! The reports $c = (1, 1, 1)$ do not lie in the image of the discrete coboundary operator B! There is a topological obstruction $\Phi_1(c) > 0$!

And what does our modern student do? He proposes levels $s = (0, -2, -1)$. He calculates the cost:

$$D_c(s) = 1|1-(-2-0)|+2|1-(-1-(-2))|+3|1-(0-(-1))| = 1(3)+2(0)+3(0) = 3$$

And then he proudly writes: 'Q.E.D.! Three is the minimum!'

An absolute scandal! A disgrace to Kharkiv! Proposing a level vector s only gives an **upper bound**: $\Phi_1(c) \leq D_c(s) = 3$. How do you know there does not exist some fractional potential $s' \in \mathbb{Q}^3$ that achieves cost $2\frac{1}{2}$? Where is your lower bound? Where is your dual witness λ? Where is your circulation?

For this examination, exhibit an exact circulation $\lambda \in \mathbb{Q}^E$ satisfying $B^T\lambda = 0$ and $|\lambda_e| \leq w_e$ whose value $\langle \lambda, c \rangle$ matches your cost 3. That is your independently checkable lower bound; failed heuristics cannot supply it!"*

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4. The Retinue and the Delegation

The room was filled with judges, each watching for a specific failure mode:

- **Liudmila (Luda)** poured steaming dark tea from the samovar into a faceted glass, sliding a saucer of wild strawberry jam toward you: > *"Breathe, young man. Don't let Dau give you heart palpitations. Drink your tea and write down the definitions cleanly: the incidence matrix B, the cochain c, the penalty weights w. It is just linear algebra over the rationals. Weak duality is your friend: $\langle \lambda, c \rangle \leq D_c(s)$. Once the two numbers meet, Dau cannot touch you."*
- **Polina** hopped up onto the edge of the dining table, dangling her sneakers and smirking: > *"Dad, ten bucks says he tries to divide by zero when he*

computes the weighted cycle formula $\min_e w_e|\sum c_e|$. *Or worse—he'll forget that edges have signs! Look at his pencil, it's shaking already."*

- **Yaroslava** reset her mechanical stopwatch with a crisp, metallic click: > *"Standard protocol. Three pages. Page 1 is elementary evaluations. Page 2 is the primal-dual furnace. Page 3 is the infinite descent trap and toy involutions. You have exactly forty-five minutes. Begemot, keep that sulfur flame away from the paper."*
- **Koroviev (Fagot)**, perched on his stool in checkered rags, beamed maliciously through his cracked pince-nez: > *"Oh, Messire! Look at his face! He wants to run 100,000 spanning trees in Python! He doesn't know that the native Rust crate* `graph_phi1.rs` *enumerates spanning forests exactly and returns* `EnumerationLimit` *when it would exceed 100,000 complete forests; the count cap is not a runtime guarantee! He thinks computation is free!"*
- **Behemoth**, the giant black cat, sat cross-legged on the table, stirring his tea with a steel compass and gnawing on a salted herring head: > *"If the candidate writes that an odd frustrated cycle can be eliminated by a local gauge transformation, I shall bite his ear off. Parity is invariant, my dear sir! You cannot turn an odd cycle into an even cycle by combing the fur!"*
- **Azazello** leaned against the window jamb, paring his yellow fingernails with an obsidian dagger: > *"One float. Just one* `0.3333`. *That is all I ask. The window is already unlatched."*
- **Abadonna** sat silently in his corner behind his opaque black goggles. If you attempted to pass an unfeasible dual circulation with $|\lambda_e| > w_e$, Abadonna would glance at your paper and turn the ink to lead poison.

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5. The Seven Mortal Sins of the Paper 001 Examinee

Landau strode to the slate and, with rapid, violent cuts of the chalk, carved the **Seven Mortal Sins of Paper 001**:

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================================================================================
            THE SEVEN MORTAL SINS OF THE PAPER 001 EXAMINEE
================================================================================
1. THE UNCERTIFIED PROPOSAL (CONFUSING UPPER BOUND WITH TRUTH):
   Asserting that a candidate potential s achieves the minimum cost without
   exhibiting an independent dual circulation lambda that certifies the lower bound.
   Penalty: Koroviev reads the candidate's diary aloud in front of the salon.

2. THE FLOATING-POINT SLOP (IEEE 754 SMUGGLING):
   Writing decimals (3.0, 0.5, 1e-6) or using gradient descent approximations.
   All assessed graph calculations use exact rationals Q; drawing conversions are not certi
   Penalty: Automatic score = -infinity; Azazello takes your pen.

3. THE OVER-CAPACITY DUAL (FEASIBILITY CHEATING):
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   Claiming a dual witness lambda that violates edge capacity (|lambda_e| > w_e)
   or fails flow conservation (B^T lambda != 0 at any vertex).
   Penalty: Behemoth overturns the boiling samovar into your lap.

4. THE INFINITE DESCENT CONFUSION (ZENO'S TRAP):
   Believing that strict rational descent (D_{t+1} < D_t) implies finite
   termination. The sequence D_t = 2^{-t} descends strictly forever without
   reaching zero.
   Penalty: Hella locks the examinee in the pantry until sunrise.

5. THE GAUGE INVARIANCE AMNESIA:
   Forgetting that shifting edge reports by an exact coboundary (c -> c + B phi)
   leaves the quotient minimum Phi_1(c) strictly unchanged.
   Penalty: Dau tosses the candidate's notebook out the third-floor window.

6. THE TOY INVOLUTION FRAUD (HYPE OVERREACH):
   Promoting the toy sign involution sigma^2 = id in InvolutionLaws.lean
   into a general solution to the continuous quantum fermion sign problem.
   Penalty: Abadonna lowers his black glasses by 2 millimeters.

7. THE 2-CELL HALLUCINATION (IMAGINARY FACES):
   Writing face boundary operators d_1 or claiming nonzero face curvature
   on a graph with zero filled 2-cells (C^2 = 0).
   Penalty: Summary expulsion to Patriarch's Ponds.
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6. The Examination Rules

Jeremy Carroll tapped his pencil on the table:

> *"Here are the ground rules for Paper 001:
>
> 1. **The Graph:** Vertices V, directed edges E, reports $c \in \mathbb{Q}^E$, penalties $w \in \mathbb{Q}^E_{>0}$.
> 2. **The Residual:** For potential $s \in \mathbb{Q}^V$, $r = c - Bs$. Cost $D_c(s) = \sum_e w_e |r_e|$.
> 3. **The Dual:** Flow $\lambda \in \mathbb{Q}^E$, conservation $B^T\lambda = 0$, capacities $|\lambda_e| \leq w_e$.
> 4. **The Zero Duality Gap:** If the dual is feasible and $\langle \lambda, c \rangle = D_c(s)$, the minimum is certified: $\Phi_1(c) = \langle \lambda, c \rangle = D_c(s)$.
> 5. **Exactness:** Every fraction must be written as p/q with $p, q \in \mathbb{Z}$ in lowest terms."*

Woland raised his silver cane and brought it down with a thunderous strike upon the hardwood floor:

"Turn the page, examinee. Let us see if you can repair a graph without drowning in floating-point fiction."

7. Examiner standard from the short companion

Landau points at the declared carrier. Woland asks which claim its checker can establish. Behemoth asks for the smallest valid near miss. Their task in this companion is **exact rational repair and dual witnesses**.

A lower-case wire token names a role only after its interpretation is declared. Arrows sketch graph edges; they do not automatically attach higher-dimensional cells. A receipt verifies its stated finite check and grants no authority to extend Genesis or make a physical claim.

The examiner's standard is a concrete calculation, an explicit counterexample, or a source-bound missing obligation. The manuscript source reading governs the answer.

The Theoretical Minimum Study Guide: Paper 001

Discrete Cochain Retraction, Primal-Dual Obstruction, and Toy Involutions

AuthorityEffect: None.
Presiding Tribunal: The Woland Retinue & Lev Davidovich Landau ("Dau").
Author of Baseline Paper: Jeremy H. Carroll.
Location: Moscow, Sadovaya 302-bis, Apartment 50.
Companion Documents: - Staging and Tribunal Lore: 00_preface_at_apartment_50.md - Examination Paper: quiz.md - Core Paper: paper.md

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0. The Landau Warning: Why Proposing Is Not Proving

Before you sit for the examination in quiz.md, you must discard the most pervasive and dangerous fallacy in modern computational science: **confusing an optimization proposal with an exact certificate**.

Professor Landau stands before the blackboard, snapping his chalk against the slate:

> *"Look at how an amateur approaches an optimization problem! He takes a function, guesses an input vector, runs twenty iterations of an approximation heuristic, and when the objective value stops moving, he proudly declares: 'The minimum is 3!'
>
> That is not mathematics; that is astrology. A proposed candidate vector gives you only an **upper bound** on the minimum: $\Phi \leq D(s)$. If you tell me you found a potential that costs 3, you have established that the minimum cannot exceed 3. But how do you know there is not another potential—perhaps fractional, perhaps subtle—that achieves cost 2.99 or 1?
>
> Heuristic failure does not prove a minimum. In this examination, construct an independent **lower-bound witness**: an object from the dual space that puts an immovable geometric floor underneath the cost function. When your upper bound from the primal proposal matches your lower bound from the dual certificate, the duality gap collapses to zero. The matching bounds prove optimality by weak duality. A complete exact search can also prove a minimum when a theorem justifies its search set; the native spanning-forest solver uses that route. Here, dual witnesses make the lower bound independently checkable."*

This study guide develops the exact discrete geometry, linear algebra, and combinatorial graph theory needed to pass the Paper 001 Theoretical Minimum.

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1. The Discrete Graph Model

1.1 Vertices, Edges, and Cochains

Let $G = (V, E)$ be a finite, connected or disconnected directed graph without self-loops. - **Vertices (0-cells):** Finite set $V = \{v_1, v_2, \dots, v_n\}$ with $|V| = n$. - **Edges (1-cells):** Finite set $E = \{e_1, e_2, \dots, e_m\}$ with $|E| = m$. Each directed edge $e = u \to v$ has source $u \in V$ and target $v \in V$ with $u \neq v$. - **0-cochains $C^0(G; \mathbb{Q})$:** Rational functions assigning a height or level to each vertex: $s : V \to \mathbb{Q}$, represented as a column vector $s \in \mathbb{Q}^V$. - **1-cochains $C^1(G; \mathbb{Q})$:** Rational functions assigning a value to each directed edge: $c : E \to \mathbb{Q}$, represented as a column vector $c \in \mathbb{Q}^E$. - **Absence of 2-cells ($C^2 = 0$):** The graph contains no filled polygonal faces. Therefore, there is no discrete 2-form curvature $d_1 c$. All obstruction in Paper 001 lives entirely in 1-cycles (the 1-homology $H_1(G; \mathbb{Q})$).

1.2 The Discrete Boundary and Coboundary

The geometry of G is governed by the edge-by-vertex incidence matrix $B \in \{-1, 0, 1\}^{|E| \times |V|}$. For each directed edge $e = u \to v$ and each vertex $x \in V$:

$$B_{e,x} = \begin{cases} -1 & \text{if } x = u \text{ (source)}, \\ +1 & \text{if } x = v \text{ (target)}, \\ 0 & \text{otherwise.} \end{cases}$$

When B acts on a vertex potential vector $s \in \mathbb{Q}^V$, it computes the directed difference across each edge:

$$(Bs)_{u \to v} = s_v - s_u$$

The transpose $B^T \in \{-1, 0, 1\}^{|V| \times |E|}$ acts as the discrete divergence operator on edge flows $\lambda \in \mathbb{Q}^E$:

$$(B^T \lambda)_x = \sum_{e = u \to x} \lambda_e - \sum_{e = x \to v} \lambda_e$$

which computes the net inflow into vertex x.

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2. The Two Costs: Proposal Cost vs. Quotient Minimum

In Paper 001, we are given: 1. **Edge Reports** $c \in \mathbb{Q}^E$**:** Inconsistent pairwise measurements (e.g. clock offsets, height differences, ranking advantages). 2. **Penalties / Weights** $w \in \mathbb{Q}^E_{>0}$**:** Strictly positive rational penalties for disagreeing with each edge report.

2.1 The Proposal Cost $D_c(s)$

Given a proposed vertex potential $s \in \mathbb{Q}^V$, the **residual 1-cochain** $r \in \mathbb{Q}^E$ measures the discrepancy between the report c and the exact potential difference Bs:

$$r = c - Bs \in \mathbb{Q}^E, \qquad r_{u\to v} = c_{u\to v} - (s_v - s_u)$$

The **proposal cost** $D_c(s)$ is the weighted ℓ_1 norm of the residual:

$$D_c(s) = \sum_{e\in E} w_e |r_e| = \sum_{e=u\to v} w_e |c_{u\to v} - (s_v - s_u)|$$

Key properties of $D_c(s)$: - $D_c(s) \geq 0$ for all $s \in \mathbb{Q}^V$. - If G has connected components $G_1, \ldots, G_k$, shifting s by an additive constant on any component leaves Bs, and hence $D_c(s)$, completely invariant. Potentials have an irrelevant gauge freedom $\mathbb{Q}^k$.

2.2 The Quotient Minimum $\Phi(c)$

The **quotient minimum** $\Phi(c)$ (also denoted $\Phi_1(c)$) is the intrinsic topological obstruction of the fixed edge reports c:

$$\Phi(c) = \min_{s\in\mathbb{Q}^V} D_c(s) = \min_{s\in\mathbb{Q}^V} \sum_{e\in E} w_e |c_e - (Bs)_e|$$

- $D_c(s)$ is a property of a **specific candidate proposal** s.
- $\Phi(c)$ is a property of the **fixed edge dataset** c.
- By definition, for every proposed potential s:

$$\Phi(c) \leq D_c(s)$$

A proposal s provides an upper bound on $\Phi(c)$.

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3. Gauge Invariance under Coboundary Shifts

Theorem: Exact Coboundary Invariance

Let $\phi \in \mathbb{Q}^V$ be any vertex potential. If we shift the edge reports c by the exact coboundary $B\phi$:

$$c' = c + B\phi$$

then the quotient minimum is strictly invariant:

$$\Phi(c') = \Phi(c)$$

Proof of Exact Coboundary Invariance

For any test potential $s \in \mathbb{Q}^V$:

$$D_{c'}(s) = \sum_e w_e |c'_e - (Bs)_e| = \sum_e w_e |(c_e + (B\phi)_e) - (Bs)_e| = \sum_e w_e |c_e - (B(s-\phi))_e|$$

Setting $f = s - \phi$, as s ranges over all of $\mathbb{Q}^V$, f also ranges over all of $\mathbb{Q}^V$. Therefore:

$$\min_{s \in \mathbb{Q}^V} D_{c'}(s) = \min_{f \in \mathbb{Q}^V} D_c(f) = \Phi(c)$$

■

Consequence: Edge reports that differ only by a coboundary belong to the same cohomology class in $C^1 / \operatorname{im} B$. For the same graph and weights, they have the same mathematical quotient minimum. Any physical interpretation needs a separately justified modeling map.

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4. Primal-Dual Witness Certification

How do we certify that a candidate potential s^* is globally optimal without exhaustive search over all of $\mathbb{Q}^V$? We construct a **dual circulation witness**.

4.1 The Dual Linear Program

The primal problem is:

$$\min_{s \in \mathbb{Q}^V} \sum_{e \in E} w_e |c_e - (Bs)_e|$$

We introduce the dual variable $\lambda \in \mathbb{Q}^E$, which represents an edge flow or circulation.

Definition: Feasible Dual Witness

A vector $\lambda \in \mathbb{Q}^E$ is a **feasible dual witness** if and only if it satisfies two conditions: 1. **Flow Conservation (Circulation Condition):** At every vertex $x \in V$, the net divergence is zero:

$$B^T \lambda = \mathbf{0} \iff \sum_{e=u \to x} \lambda_e = \sum_{e=x \to v} \lambda_e \quad \forall x \in V$$

2. **Capacity Bounds:** On every edge $e \in E$, the flow does not exceed the edge penalty:

$$|\lambda_e| \leq w_e \quad \forall e \in E$$

The **dual objective value** is the standard inner product of the circulation with the reports:

$$\langle \lambda, c \rangle = \sum_{e \in E} \lambda_e c_e$$

4.2 Theorem: Weak Duality (The Iron Floor)

For **any** proposed vertex potential $s \in \mathbb{Q}^V$ and **any** feasible dual witness $\lambda \in \mathbb{Q}^E$:

$$\langle \lambda, c \rangle \leq D_c(s)$$

Proof of Weak Duality

Since $B^T \lambda = \mathbf{0}$, taking the dot product with the potential vector s gives:

$$s^T (B^T \lambda) = 0 \iff (Bs)^T \lambda = \sum_{e \in E} (Bs)_e \lambda_e = 0$$

Therefore, we can subtract zero without changing the value:

$$\langle \lambda, c \rangle = \sum_{e \in E} \lambda_e c_e - \sum_{e \in E} \lambda_e (Bs)_e = \sum_{e \in E} \lambda_e (c_e - (Bs)_e) = \sum_{e \in E} \lambda_e r_e$$

Applying the triangle inequality and the capacity bound $|\lambda_e| \leq w_e$:

$$\langle \lambda, c \rangle = \sum_{e \in E} \lambda_e r_e \leq \sum_{e \in E} |\lambda_e||r_e| \leq \sum_{e \in E} w_e |r_e| = D_c(s)$$

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4.3 Theorem: Certificate of Optimality (Zero Duality Gap)

If a proposer exhibits a candidate potential $s^* \in \mathbb{Q}^V$ and a referee exhibits a feasible dual witness $\lambda^* \in \mathbb{Q}^E$ such that:

$$\langle \lambda^*, c \rangle = D_c(s^*)$$

then s^* is **certified by weak duality to be an exact global minimizer**, and the quotient minimum is:

$$\Phi(c) = D_c(s^*) = \langle \lambda^*, c \rangle$$

Proof of Optimality Certificate

By weak duality, for every potential $s \in \mathbb{Q}^V$:

$$D_c(s) \geq \langle \lambda^*, c \rangle = D_c(s^*)$$

Thus no potential can achieve a cost strictly less than $D_c(s^*)$.

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5. The Manuscript Triangle: Complete Worked Example

Let us work through the foundational test case of Paper 001: the **Manuscript Triangle**.

5.1 The Graph Specification

- **Vertices:** $V = \{a, b, c\}$.
- **Edges:** Three directed edges forming a cycle:
  - $e_1 = a \to b$
  - $e_2 = b \to c$
  - $e_3 = c \to a$

- **Reports:** $c = (c_1, c_2, c_3) = (1, 1, 1)$.
- **Weights:** $w = (w_1, w_2, w_3) = (1, 2, 3)$.

The incidence matrix $B \in \{-1, 0, 1\}^{3\times 3}$ is:

$$B = \begin{pmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \\ 1 & 0 & -1 \end{pmatrix}$$

The cycle sum of reports is:

$$\sum_{e\in\text{cycle}} c_e = c_1 + c_2 + c_3 = 1 + 1 + 1 = 3 \neq 0$$

Because the cycle sum is nonzero, no potential s can make all residuals zero. There is an intrinsic obstruction.

5.2 Evaluating Candidate Proposals

Candidate 1: The All-Zero Potential $s^{(0)} = (0, 0, 0)$

- Differences: $Bs^{(0)} = (0-0, 0-0, 0-0) = (0, 0, 0)$.
- Residual: $r = c - Bs^{(0)} = (1, 1, 1)$.
- Cost:
$$D_c(s^{(0)}) = 1|1| + 2|1| + 3|1| = 1 + 2 + 3 = 6$$

Candidate 2: The Optimal Potential $s^* = (0, -2, -1)$

- Differences:
  - $(Bs^*)_{a\to b} = s_b - s_a = -2 - 0 = -2$
  - $(Bs^*)_{b\to c} = s_c - s_b = -1 - (-2) = 1$
  - $(Bs^*)_{c\to a} = s_a - s_c = 0 - (-1) = 1$
- Residuals:
  - $r_1 = c_1 - (s_b - s_a) = 1 - (-2) = 3$
  - $r_2 = c_2 - (s_c - s_b) = 1 - 1 = 0$
  - $r_3 = c_3 - (s_a - s_c) = 1 - 1 = 0$

$$r = (3, 0, 0)$$

- Cost:

$$D_c(s^*) = w_1|r_1| + w_2|r_2| + w_3|r_3| = 1(3) + 2(0) + 3(0) = 3$$

We now have an upper bound: $\Phi(c) \leq 3$.

5.3 Constructing the Dual Witness λ^*

We require a circulation $\lambda = (\lambda_1, \lambda_2, \lambda_3)$ such that: 1. Flow conservation at each vertex: - At a: incoming e_3, outgoing $e_1 \implies \lambda_3 - \lambda_1 = 0 \implies \lambda_1 = \lambda_3$. - At b: incoming e_1, outgoing $e_2 \implies \lambda_1 - \lambda_2 = 0 \implies \lambda_2 = \lambda_1$. - At c: incoming e_2, outgoing $e_3 \implies \lambda_2 - \lambda_3 = 0$. Therefore, any circulation on a simple directed cycle must have uniform flow:

$$\lambda = (k, k, k) \quad \text{for some } k \in \mathbb{Q}$$

2. Capacity constraints $|\lambda_e| \leq w_e$: - $|\lambda_1| \leq 1 \implies |k| \leq 1$ - $|\lambda_2| \leq 2 \implies |k| \leq 2$ - $|\lambda_3| \leq 3 \implies |k| \leq 3$ The bottleneck capacity is the **minimum edge weight**:

$$|k| \leq \min(1, 2, 3) = 1$$

To maximize the dual objective $\langle \lambda, c \rangle = k(1) + k(1) + k(1) = 3k$, we choose the largest feasible flow:

$$k = 1 \implies \lambda^* = (1, 1, 1)$$

3. Evaluate the Dual Objective:

$$\langle \lambda^*, c \rangle = 1(1) + 1(1) + 1(1) = 3$$

4. Duality Gap Verification:

$$D_c(s^*) - \langle \lambda^*, c \rangle = 3 - 3 = 0$$

The duality gap is zero! The candidate $s^* = (0, -2, -1)$ is mathematically certified as an exact global minimizer, and the true quotient obstruction is $\Phi(c) = 3$.

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6. The Directed Cycle Formula

The result on the Manuscript Triangle generalizes to any single directed cycle.

Theorem: Single Directed Cycle Obstruction

Let $C = (V_C, E_C)$ be a simple directed cycle with edge reports c_e and positive penalties $w_e > 0$. Then:

$$\Phi(c) = \left(\min_{e \in E_C} w_e \right) \left| \sum_{e \in E_C} c_e \right|$$

Proof of Directed Cycle Formula

1. **Lower Bound (Dual):** Any circulation λ on a simple cycle must be a constant flow along the cycle: $\lambda_e = k$ for all $e \in E_C$. Capacity requires $|\lambda_e| \leq w_e$ for all $e \in E_C$, which forces:

$$|k| \leq \min_{e \in E_C} w_e$$

The dual value is:

$$\langle \lambda, c \rangle = \sum_{e \in E_C} k c_e = k \sum_{e \in E_C} c_e$$

Choosing $k = \operatorname{sgn}\left(\sum_{e \in E_C} c_e\right) \min_{e \in E_C} w_e$, we obtain the lower bound:

$$\Phi(c) \geq \left(\min_{e \in E_C} w_e\right) \left|\sum_{e \in E_C} c_e\right|$$

2. **Upper Bound (Primal):** Let $e^* = u^* \to v^*$ be the bottleneck edge that achieves the minimum penalty: $w_{e^*} = \min_{e \in E_C} w_e$. Remove e^* from the cycle. The remaining edges $E_C \setminus \{e^*\}$ form a spanning tree (a path). On any tree, edge reports can be matched exactly by a vertex potential ($r_e = 0$ for all $e \neq e^*$). Solving for the potential on the path dumps the entire cycle discrepancy onto the single bottleneck edge e^*:

$$|r_{e^*}| = \left|\sum_{e \in E_C} c_e\right|$$

The cost of this potential is:

$$D_c(s^*) = w_{e^*} |r_{e^*}| = \left(\min_{e \in E_C} w_e\right) \left|\sum_{e \in E_C} c_e\right|$$

Since the upper bound equals the lower bound, the formula is exact.

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7. Sparsity and Uniqueness

7.1 Sparsity of Optimal Residuals

In linear programming over polyhedra, extreme points (basic feasible solutions) exhibit structural sparsity.

Theorem: Residual Support Bound For any finite graph $G = (V, E)$, there exists an optimal potential s^* whose residual $r^* = c - Bs^*$ has support size bounded by:

$$|\operatorname{supp}(r^*)| \leq |E| - \operatorname{rank}(B)$$

Since $\operatorname{rank}(B) = |V| - k$ (where k is the number of connected components):

$$|\operatorname{supp}(r^*)| \leq |E| - |V| + k = \dim H_1(G; \mathbb{Q})$$

The number of nonzero residual edges in a basic optimal solution is at most the **first Betti number** (the cycle rank of the graph).

- For the Manuscript Triangle: $|E| = 3$, $|V| = 3$, $k = 1 \implies \dim H_1 = 3 - 3 + 1 = 1$.
- Indeed, the optimal residual $r^* = (3, 0, 0)$ has $|\operatorname{supp}(r^*)| = 1 \leq 1$.

7.2 Uniqueness and the Weighted Null-Space Property

A potential always has additive freedom on each connected component. Uniqueness here means a unique residual, or a unique potential after one vertex per component has been fixed.

Let $S = \operatorname{supp}(r^*)$. The relevant residual perturbations lie in $\operatorname{im} B$, because changing a potential changes a residual by a vector Bh. A sufficient condition is

$$\sum_{e \in S} w_e |z_e| < \sum_{e \notin S} w_e |z_e| \qquad \text{for every } 0 \neq z \in \operatorname{im} B.$$

Then any distinct feasible residual $r^* + z$ satisfies

$$\|r^* + z\|_{1,w} \geq \|r^*\|_{1,w} - \|z_S\|_{1,w} + \|z_{S^c}\|_{1,w} > \|r^*\|_{1,w}.$$

This condition is sufficient for the particular residual. Its failure alone does not prove nonuniqueness. The dual witnesses instead lie in $\ker(B^T)$; these two spaces serve different roles.

On the equal-weight triangle with reports $(1, 1, 1)$ and weights $(2, 2, 2)$, the residuals $(3, 0, 0)$ and $(0, 3, 0)$ both cost 6. They correspond to potentials $(0, -2, -1)$ and $(0, 1, -1)$. The feasible dual $(2, 2, 2)$ has objective 6, proving both are optimal. For $S = \{e_1\}$, the cut vector $z = (1, -1, 0) = B(0, 1, 0)$ gives equality in the two weighted sums, so the sufficient condition fails.

A useful error detector: every circulation on this directed triangle is $t(1,1,1)$. Substituting circulations into the displayed condition would give $2|t| < 4|t|$ and incorrectly predict uniqueness. Always check the perturbation space.

7.3 A sparse optimum does not make every optimum sparse

With equal weights and positive cycle sum h, every same-sign residual split with $\sum r_e = h$ is optimal. On the triangle this is a two-dimensional simplex. It contains the three single-edge residuals and the dense residual $(1,1,1)$. Thus cycle rank 1 bounds the support of **some** optimum, while the optimal face can have dimension 2.

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8. Halting Pitfalls and Zeno's Trap

8.1 The Counterexample: Strict Descent Without Termination

Consider a single edge with report $c = 1$, weight $w = 1$. Suppose an optimization routine generates a sequence of potential differences such that the cost at step t is:

$$D_t = 2^{-t}$$

Notice: 1. **Strict Monotonic Descent:** At every single step, $D_{t+1} = 2^{-(t+1)} < 2^{-t} = D_t$. 2. **Infinite Rational Sequence:** The sequence strictly decreases for all $t \in \mathbb{N}$ without ever reaching the infimum 0 at any finite t.

The Cardinal Halting Law:

> **Strict rational descent alone does NOT imply finite termination.**

To make a stopping argument, state which quantity is bounded and which calls must return:

- If all costs lie on a fixed lattice $\delta\mathbb{Z}$ with $\delta > 0$, are bounded below, and each admitted step strictly decreases cost, there are finitely many admitted descents from a finite starting cost. Rejected attempts still need their own limit.
- A finite exhaustive search terminates if it visits its finite candidate set without repetition and every candidate check returns. Searching extreme points without these conditions is not a halting proof.
- A finite fuel wrapper bounds attempted transitions. A timeout or a proved bound is separately needed for each parser, solver, or external call. Fuel exhaustion means an unfinished computation, not convergence.

In the supplied `SymbolGameGovernor`, `turn` counts calls; the class has no fuel limit. Demo 1 enforces fuel in its outer `for` loop and reports `UNFINISHED` while cost is still positive. There is no `FUEL_EXHAUSTED` member in the governor's verdict enum. The native adapter has a separate 30-second subprocess timeout. Its forest cap counts complete forests, not every recursive branch or elapsed time.

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9. Software Implementations and the 100,000 Forest Cap

Paper 001 provides two reference implementations:

9.1 The Native Rust Solver (`code/rust/src/graph_phi1.rs`)

- **Algorithm:** Bounded exact exhaustive enumeration over spanning forests.
- **Safety Ceiling:** Capped strictly at 100,000 forests.
- **Behavior:**
  - If the graph has $\leq 100,000$ spanning forests, it computes the exact rational minimum $\Phi_1(c)$ and returns `phi1`, `gauge`, and `forests_checked`.
  - If the search exceeds 100,000 forests, it **refuses when the next complete forest would exceed the cap**, with typed error `EnumerationLimit`. It does **not** return a partial or approximate minimum.
- **Limitation:** Exponential in graph size; returns the primal potential, but does not construct the dual circulation certificate.

9.2 The Paper-Local Python Reference (`code/symbol_game_governor.py`)

- Constructs both the primal potential s^* and the dual circulation λ^*.
- Independently verifies the zero duality gap: $\langle \lambda^*, c \rangle = D_c(s^*)$.
- Implements a governor whose `CUT` admission rule requires strict cost descent. `ASSERT` changes reports and `RETRACT` restores a state; neither is covered by the fixed-report descent rule.

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10. Lean 4 Toy Involutions and Claim Hygiene

The manuscript describes these narrowly scoped local statements:

1. `OddCycleObstruction.lean`: no assignment of signs ± 1 to the three vertices of a triangle makes every edge product equal to -1.
2. `InvolutionLaws.lean`: a configuration and its image under a supplied sign-reversing, weight-preserving map have signed natural-number weights summing to zero. The map and its hypotheses are supplied.

3. `NonAuthority.lean`: a declared policy is encoded as a type with one value. Its theorem holds by construction and checks no runtime behavior.

These descriptions do not establish a general graph LP, production implementation correctness, termination of arbitrary rational descent, a polynomial-time algorithm, or a physical fermion sign-problem solution. Read the actual theorem statements, pinned toolchain, and axiom report before claiming a successful formal replay. Those files must travel with the deposit.

To bridge a finite cochain model to a physical representation, specify the physical carrier, the representation, the map to the graph data, and a theorem stating exactly what signs, weights, or observables the map preserves. Any runtime or complexity claim needs its own correspondence proof and cost model.

11. Triangle recap and complete instructor key

These lines recap the Manuscript Triangle. The worked derivation is in section 5. The separate instructor key covers all thirteen examination problems, a 100-point rubric, and the optional final challenges.

1. The coefficients are rational, the edge order is ab, bc, ca, and this graph has no declared 2-cells. A displayed cycle does not attach a face.
2. For $f = (0, -2, -1)$, $df = Bs^* = (-2, 1, 1)$, the residual is $(3, 0, 0)$, and the weighted cost is 3. A feasible potential gives an upper bound on the minimum.
3. $\lambda = (1, 1, 1)$ has zero divergence and satisfies $|\lambda| \leq w$ for $w = (1, 2, 3)$. Its objective is 3, matching the candidate. With $w = (2, 2, 2)$, $\lambda = (2, 2, 2)$ gives objective 6 and the same f has cost 6. The answer 2 is impossible.
4. Translating the potential variable leaves the quotient minimum unchanged. Replacing c with df changes its cohomology class when the original cycle sum is nonzero; that is a different instance, not gauge erasure.
5. A physical modeling bridge, an independently checked carrier interpretation, and any requested complexity theorem are still owed. No general fermion sign-problem solution follows.

The manuscript section A certificate for optimality supplies the coefficient system and move grammar. The package deposit verifier checks file custody rather than the truth of every answer. The standalone companion needs no external worksheet to define its assessed graph model. If the shared repository worksheet is included later, use a release-pinned link and verify it from the exported package.

Theoretical Minimum Examination: Paper 001

AuthorityEffect: None.
Presiding Tribunal: The Woland Retinue & Lev Davidovich Landau ("Dau").
Author of Baseline Paper: Jeremy H. Carroll.
Location: Moscow, Sadovaya 302-bis, Apartment 50.
Companion Preface: Read 00_preface_at_apartment_50.md for tribunal staging, character stations, and the Seven Mortal Sins.
Study Guide: Master study_guide.md before sitting for this examination.

> [!IMPORTANT] **Instructions from Professor Landau:** 1. All calculations must be performed in exact rationals $\mathbb{Q} = \{p/q : p \in \mathbb{Z}, q \in \mathbb{Z}_{>0}\}$. 2. Zero floating-point numbers, zero decimals, zero heuristic approximations. 3. An optimization proposal is not a theorem. For this examination, certify every calculated minimum with a feasible dual witness. Complete exact enumeration can also prove a minimum when its search-set theorem and exhaustion are established. 4. Work through the pages in order. Attempt every question and use feedback to repair earlier errors. The complete instructor key has hints, worked answers, and a 100-point rubric; a perfect paper earns the tribunal seal.

Page 1: Horribly Simple (The Kindergarten of Inconsistent Offsets)

Landau tosses his chalk: "If you cannot evaluate the cost of three edges on a napkin, you belong in a business school, not Apartment 50."

Problem 1.1: The Three Inconsistent Clocks

Three instruments report relative time offsets between three nodes a, b, c: - Instrument 1 reports: $s_b - s_a = 1$ (directed edge $e_1 = a \to b$, report $c_1 = 1$). - Instrument 2 reports: $s_c - s_b = 1$ (directed edge $e_2 = b \to c$, report $c_2 = 1$). - Instrument 3 reports: $s_a - s_c = 1$ (directed edge $e_3 = c \to a$, report $c_3 = 1$).

1. Compute the sum of the reports around the directed cycle: $\sum_{i=1}^{3} c_i$.
2. Explain in one sentence why no potential assignment $s = (s_a, s_b, s_c) \in \mathbb{Q}^3$ can satisfy all three reports simultaneously.
3. Does the graph contain any 2-cells (filled faces)? State the dimension of $C^2(G; \mathbb{Q})$.

Problem 1.2: The All-Zero Potential

Let the edge penalties for disagreeing with the reports be $w = (w_1, w_2, w_3) = (1, 2, 3)$. Consider the all-zero candidate potential $s^{(0)} = (0, 0, 0)$. 1. Compute the discrete coboundary vector $Bs^{(0)} \in \mathbb{Q}^3$. 2. Compute the residual vector $r^{(0)} = c - Bs^{(0)}$. 3. Calculate the exact proposal cost:

$$D_c(s^{(0)}) = \sum_{i=1}^{3} w_i |r_i^{(0)}|$$

Problem 1.3: The Candidate Proposal $s = (0, -2, -1)$

A candidate proposes the vertex potential $s^* = (s_a, s_b, s_c) = (0, -2, -1)$. 1. Calculate the potential difference across each edge: - $(Bs^*)_{a\to b} = s_b - s_a =$ _____ - $(Bs^*)_{b\to c} = s_c - s_b =$ _____ - $(Bs^*)_{c\to a} = s_a - s_c =$ _____ 2. Compute the residual 1-cochain $r^* = c - Bs^* = (r_1^*, r_2^*, r_3^*)$. 3. Compute the exact proposal cost $D_c(s^*)$.

Problem 1.4: Proposal vs. Lower Bound Certificate

A machine learning agent runs a search routine, discovers the potential $s^* = (0, -2, -1)$ with cost $D_c(s^*) = 3$, and states: > *"The cost is 3, and since three search heuristics failed to find a lower value, the global minimum is mathematically proved to be 3."*

1. What mathematical bound does $D_c(s^*) = 3$ provide on the quotient minimum $\Phi(c)$?

   Choose: $[\Phi(c) \geq 3 \quad | \quad \Phi(c) \leq 3 \quad | \quad \Phi(c) = 3]$

2. Why does heuristic failure NOT establish that $\Phi(c) = 3$?
3. What mathematical object must be exhibited to certify an independent lower bound $\Phi(c) \geq 3$?

Page 2: Intermediate Rigor (The Primal-Dual Cochain Engine)

Landau taps the slate: "Now we build the dual furnace. Anyone can guess a level; only a mathematician can construct a circulation."

Problem 2.1: Dual Feasibility Conditions

Let $G = (V, E)$ be a directed graph with incidence matrix $B \in \{-1, 0, 1\}^{|E|\times|V|}$, edge reports $c \in \mathbb{Q}^E$, and penalty weights $w \in \mathbb{Q}^E_{>0}$. A vector $\lambda \in \mathbb{Q}^E$ is a candidate dual witness. 1. State the exact linear equation that λ must satisfy

at each vertex $v \in V$ to be a valid circulation. 2. State the inequality constraint that λ_e must satisfy on each edge $e \in E$. 3. If G is a simple directed cycle of length k, show that every valid circulation λ must have constant magnitude across all edges: $\lambda_e = \alpha$ for all $e \in E$.

Problem 2.2: The Weak Duality Derivation

Derive the Weak Duality Theorem: For **every** potential $s \in \mathbb{Q}^V$ and **every** feasible dual circulation $\lambda \in \mathbb{Q}^E$:

$$\langle \lambda, c \rangle \leq D_c(s)$$

Hint: Use $B^T\lambda = \mathbf{0}$ to show that $\sum_e \lambda_e (Bs)_e = 0$, express $\langle \lambda, c \rangle$ in terms of the residual $r = c - Bs$, and apply the capacity bound $|\lambda_e| \leq w_e$. Show every step cleanly.

Problem 2.3: Zero Duality Gap on the Manuscript Triangle

Return to the Manuscript Triangle with $c = (1, 1, 1)$ and $w = (1, 2, 3)$. 1. What is the maximum value that a constant circulation $\lambda = (k, k, k)$ can attain without violating the capacity constraints $|\lambda_i| \leq w_i$? 2. Exhibit the optimal dual witness vector $\lambda^* \in \mathbb{Q}^3$. 3. Compute the dual objective value:

$$\langle \lambda^*, c \rangle = \sum_{i=1}^{3} \lambda_i^* c_i$$

4. Compute the duality gap:

$$\Delta = D_c(s^*) - \langle \lambda^*, c \rangle$$

where $s^* = (0, -2, -1)$. State the definitive conclusion regarding the global minimum $\Phi(c)$.

Problem 2.4: Coboundary Gauge Invariance

Let $c = (1, 1, 1)$ be the reports on the Manuscript Triangle. A gauge transformation $\phi = (2, -1, 5) \in \mathbb{Q}^3$ is applied to the vertices, generating new reports:

$$c' = c + B\phi$$

1. Calculate the shifted report vector $c' \in \mathbb{Q}^3$. 2. Compute the cycle sum $\sum_{i=1}^{3} c'_i$. 3. Without re-running an optimization solver, determine the exact value of $\Phi(c')$. Cite the governing theorem.

Problem 2.5: The Directed Cycle Obstruction Formula

Consider a directed 4-cycle C_4 with vertices $\{v_1, v_2, v_3, v_4\}$, directed edges $e_1 = v_1 \to v_2$, $e_2 = v_2 \to v_3$, $e_3 = v_3 \to v_4$, $e_4 = v_4 \to v_1$. The edge reports are

$c = (4, -2, 5, 1)$ and the penalties are $w = (7, 3, 9, 5)$. 1. State the general Directed Cycle Obstruction Formula for $\Phi(c)$. 2. Identify the bottleneck edge e^* and its weight $\min_e w_e$. 3. Calculate the exact quotient minimum $\Phi(c)$ as an integer or fraction in lowest terms.

Page 3: Adversarial Reality Gates & Invariant Involutions (The Descent Crucible)

Woland smiles faintly. Landau leans in: "Here is where the pretenders drown. If your algorithm descends forever without stopping, Azazello will assist your descent through the floorboards."

Problem 3.1: The Infinite Rational Descent Trap (Zeno's Trap)

Consider a single edge $e = u \to v$ with report $c = 1$ and penalty $w = 1$. An autonomous solver generates a sequence of potential differences $\Delta s_t = s_v(t) - s_u(t)$ such that the residual at step t is $r_t = 2^{-t}$ for $t = 1, 2, 3, \ldots$. 1. Compute the cost D_t at step t. Is the sequence strictly monotonically decreasing ($D_{t+1} < D_t$) for all t? 2. Does there exist any finite step $t^* \in \mathbb{N}$ such that $D_{t^*} = 0$? 3. Explain why the rule *"Admit proposal if $D(s_{new}) < D(s_{old})$"* is an unsafe halting condition in discrete optimization over $\mathbb{Q}$. 4. Separate three obligations: strict `CUT` descent, a finite attempt budget, and a returning or timed-out check. Which are enforced by `SymbolGameGovernor`, which by Demo 1's outer loop, and which by the native adapter? Explain why a turn counter alone is insufficient.

Problem 3.2: Sparsity of Optimal Residuals & The Null-Space Property

Let $G = (V, E)$ be a connected graph with n vertices and m edges ($m \geq n$). 1. State the upper bound on the number of non-zero entries in the residual vector $r^* = c - Bs^*$ for an extreme-point (basic feasible) optimal solution:

$$|\operatorname{supp}(r^*)| \leq _____$$

Express the bound in terms of the cycle rank (first Betti number) $\beta_1 = m - n + 1$. 2. Suppose G is a triangle ($n = 3, m = 3$) where all edge penalties are strictly equal: $w = (2, 2, 2)$, with reports $c = (1, 1, 1)$. - Does there exist a uniquely minimizing potential s^*? - Exhibit two distinct potentials $s^{(1)}$ and $s^{(2)}$ (not differing by an additive constant) that both achieve the global minimum $\Phi(c) = 6$. - Using $f = (0, -2, -1)$ and $\lambda = (2, 2, 2)$, show why a proposed minimum of 2 is impossible. - State the sufficient **Weighted Null-Space Property** using nonzero $z \in \operatorname{im} B$. For $S = \{e_1\}$, exhibit a vector that makes the strict inequality fail. Does failure alone prove that a particular residual is nonunique?

Problem 3.3: Combinatorial Spanning Tree Search & The 100,000 Forest Cap

The native Rust solver `graph_phi1.rs` solves Φ_1 by enumerating spanning forests over G. 1. According to Cayley's formula, the complete graph K_n has n^{n-2} spanning trees. - Calculate the number of spanning trees for K_6. - Calculate the number of spanning trees for K_8. 2. If an examinee passes K_8 to `graph_phi1.rs`, what is the exact behavior of the Rust solver? - Does it return an approximate floating-point answer? - Does it return a partial minimum based on the first 100,000 trees? - What typed enum error does it return? 3. Why does repository governance forbid the solver from returning a "best-so-far" partial minimum when the cap is hit?

Problem 3.4: Lean 4 Toy Involutions and Claim Discipline

In the repository formalization `lean4/`: The manuscript describes the following local statements; inspect the actual files before asserting a build result: - `InvolutionLaws.lean` proves pair cancellation for a supplied sign-reversing, weight-preserving map on signed natural-number weights. - `OddCycleObstruction.lean` concerns the three vertex signs of a triangle. - `NonAuthority.lean` encodes a policy type with one value, so its theorem holds by construction.

An eager machine learning researcher publishes a preprint claiming: > *"The Lean 4 proofs in Paper 001 mathematically verify a polynomial-time algorithm that solves the quantum fermion sign problem."*

Write the formal **Adversarial Receipt Audit** refuting this claim according to the repository Claim Hygiene Rule: 1. **What it supports:** State precisely what `OddCycleObstruction.lean` and `InvolutionLaws.lean` actually formalize in this toy setting. 2. **What it does NOT support:** State three specific physical and computational claims that do NOT follow from these toy Lean files. 3. **The narrowest defensible claim:** Formulate the single narrowest valid claim regarding the Lean 4 formalization in Paper 001. 4. **The next test needed:** Specify the carrier and representation, the mapping theorem, and any additional algorithm or complexity theorem needed for the proposed physical claim. Explain why a toy theorem or a hash receipt cannot supply these obligations.

Grading Trichotomy & Terminal Disposition

Use the problem weights in instructor_key.md. Grade all attempted answers and give a concrete repair for each lost point. These dispositions are fictional game labels, not judgments about a student.

Landau dusts his hands, spins his railway chalk, and glares across the table:

- `PASS` **(Score: 100/100):** Every numerical quantity is in exact rationals $\mathbb{Q}$; all dual circulation witnesses are exhibited with zero duality gap; the weak duality theorem is derived cleanly without missing steps; Cayley's formula and the 100,000 cap are evaluated accurately; and the Lean 4 claim hygiene audit is uncompromisingly strict. *Disposition:* Woland smiles; Landau signs the slate; Jeremy Carroll shakes your hand; Liudmila pours a celebratory cup of tea with strawberry jam.
- `NONDESCENT` **(Score: 40–99/100):** The student confused an upper bound with a certificate, made arithmetic errors on the cycle formula, or handwaved the basic feasible solution sparsity bound. *Disposition:* Koroviev cackles shrilly; the candidate is sent to the kitchen to compute tree enumerations by hand under Polina's supervision.
- `DECAPITATED` **(Score: 0–39/100):** The student used floating-point decimals, asserted that 2^{-t} terminates in finite time, or claimed that toy Lean involutions solve the quantum fermion sign problem. *Disposition:* Azazello points to the open window; Behemoth ignites the primus; Abadonna removes his glasses.

Final challenges in Apartment 50

These optional challenges come after the examination. They do not change its 100-point score. Woland wants a proof; Behemoth wants a counterexample. Hints and solutions are in the instructor key.

Challenge A The sufficient condition that fails

Orient the six edges of K_4 from the earlier letter to the later letter, with vertex order a, b, c, d, and give every edge weight 1. Set $c_{ab} = 1$, $c_{ac} = -1$, and all other reports to 0.

1. Show that $s = 0$ has cost 2. Verify the dual witness, in edge order ab, ac, ad, bc, bd, cd,

$$\lambda = (1, -1, 0, 1/2, 1/2, -1/2).$$

2. Prove that the minimizing potential is unique after fixing $s_a = 0$. Use the strict capacities on the four zero-residual edges.
3. For $S = \{ab, ac\}$ and $h = (1, 0, 0, 0)$, compute $z = Bh$. Show that the sufficient weighted null-space condition fails despite the unique minimizer.
4. State exactly why this does not contradict the theorem in the paper.

Challenge B The perfect-looking counterfeit receipt

The equal-weight triangle has reports $(1, 1, 1)$ and weights $(2, 2, 2)$. A native response claims `PASS`, potential $(0, -2, -1)$, and minimum `2`.

1. Recompute the potential's cost. Which consistency check rejects the response before optimality is considered?

2. Give a feasible dual that excludes a minimum of 2.
3. Write a response-validation checklist covering JSON shape, exact rational strings, complete vertex coverage, forest-count type and range, and objective agreement.
4. Does a structurally valid response and a matching primal cost prove global minimality? Name the remaining trust or certificate obligation. What should happen on a native refusal or timeout?

Paper 001 instructor key and final challenges

Proposed companion for the v0.1 preprint, reviewed 2026-10-02. This key follows the corrected quiz supplied with this proposal. It does not attest to an unavailable Rust or Lean build. The tribunal dialogue is fictional.

Students should attempt each problem before consulting its answer. Give feedback on all attempted problems; an early error should identify what to revisit, rather than conceal the later learning material. Use exact integers and fractions throughout the assessed calculations.

Learning route and rubric

Problem	Idea to learn	Points
1.1	Telescoping and the declared cell carrier	6
1.2	Evaluate a proposal cost	5
1.3	Distinguish a fitted difference from a residual	6
1.4	Upper bounds and missing lower bounds	7
2.1	Balance and edge capacities	7
2.2	Prove weak duality	12
2.3	Verify a matching pair of witnesses	7
2.4	Preserve the quotient under a report shift	6
2.5	Construct a cycle optimum	8
3.1	Separate descent, fuel, and time bounds	9
3.2	Separate sparse existence, uniqueness, and gauge freedom	12
3.3	Exact enumeration and honest refusal	7
3.4	Read the scope of a formal theorem	8
Total		100

Award arithmetic credit for correct exact computations and proof credit for stated hypotheses and justified steps. For multi-part problems, divide the allotted points among the parts; put most of 2.2's points on the two justified equalities and the capacity inequality. The existing perfect-score tribunal seal can remain a game reward. A lower score should come with specific repairs and another attempt. The two final challenges are optional and unscored.

Page 1 worked answers

Problem 1.1

The report sum is 3. Potential differences telescope around the cycle to 0, so they cannot equal reports whose sum is 3. No 2-cells are declared; $C^2 = \{0\}$ has dimension 0. Drawing the triangle does not fill it.

Hint for a stalled student: write $(s_b - s_a) + (s_c - s_b) + (s_a - s_c)$ before computing anything else.

Problem 1.2

$Bs^{(0)} = (0, 0, 0)$, $r^{(0)} = (1, 1, 1)$, and $D_c(s^{(0)}) = 1 + 2 + 3 = 6$.

Problem 1.3

$Bs^* = (-2, 1, 1)$, $r^* = (3, 0, 0)$, and $D_c(s^*) = 1 \cdot 3 + 2 \cdot 0 + 3 \cdot 0 = 3$. The negative fitted difference on the first edge is allowed; it is the residual, not the potential sign, that incurs cost.

Problem 1.4

The candidate proves $\Phi(c) \leq 3$. Failed heuristic searches provide no bound on unvisited potentials. For the requested independent certificate, exhibit a circulation λ with $B^T\lambda = 0$, $|\lambda_e| \leq w_e$, and $\lambda^T c = 3$. The witness $(1, 1, 1)$ works. Together with the candidate, it proves $\Phi(c) = 3$.

This examination asks for dual certificates. It should not teach that duality is the only possible proof: a complete exact enumeration with a proved search-set theorem can also establish a minimum.

Page 2 worked answers

Problem 2.1

At each vertex, incoming flow equals outgoing flow:

$$\sum_{e:u\to v} \lambda_e - \sum_{e:v\to u} \lambda_e = 0.$$

Equivalently $B^T\lambda = 0$. Each edge satisfies $-w_e \leq \lambda_e \leq w_e$. On a consistently oriented simple directed cycle, every vertex has one incoming and one outgoing edge, so consecutive flows are equal and $\lambda_e = \alpha$ everywhere. Capacity gives $|\alpha| \leq \min_e w_e$.

Problem 2.2

Let $r = c - Bs$. Since the dual is balanced,

$$\lambda^T Bs = s^T B^T \lambda = 0.$$

Consequently

$$\lambda^T c = \lambda^T (c - Bs) = \sum_e \lambda_e r_e \leq \sum_e |\lambda_e|\,|r_e| \leq \sum_e w_e |r_e| = D_c(s).$$

This holds for every potential and every feasible dual. Therefore a feasible dual gives a lower bound, while a potential gives an upper bound. Equal bounds

prove optimality using weak duality; no separate strong-duality invocation is needed for a supplied matching pair.

Problem 2.3

The largest feasible constant k is 1, because the least capacity is 1. Thus $\lambda^* = (1, 1, 1)$ is balanced and within all capacities. Its objective is $1 + 1 + 1 = 3$. The candidate also costs 3, so the gap is 0 and $\Phi(c) = 3$. This certifies a minimum repair cost, not exact consistency of the original reports.

Problem 2.4

$B\phi = (-3, 6, -3)$, so $c' = (-2, 7, -2)$ and the cycle sum remains 3. Coboundary invariance gives $\Phi(c') = \Phi(c) = 3$ for the same weights.

For an explicit matching pair, take $s' = s^* + \phi = (2, -3, 4)$ and the unchanged dual $(1, 1, 1)$. The fitted differences are $(-5, 7, -2)$, the residual is $(3, 0, 0)$, and both objectives equal 3.

Problem 2.5

For a consistently directed cycle,

$$\Phi(c) = \rho|h|, \qquad \rho = \min_e w_e, \quad h = \sum_e c_e.$$

Here $h = 4 - 2 + 5 + 1 = 8$, $\rho = 3$, and the bottleneck is e_2. Hence $\Phi(c) = 24$.

To meet the certificate instruction, use potential $(s_{v_1}, s_{v_2}, s_{v_3}, s_{v_4}) = (0, 4, -6, -1)$. It gives $Bs = (4, -10, 5, 1)$ and residual $(0, 8, 0, 0)$, costing 24. The constant circulation $(3, 3, 3, 3)$ is feasible and has objective $3(8) = 24$.

Page 3 worked answers

Problem 3.1

$D_t = 2^{-t}$ and $D_{t+1} = D_t/2 < D_t$. No finite t has $D_t = 0$. In fact the single edge admits an exact potential difference 1, so the minimum is 0; the chosen sequence merely fails to reach it.

The obligations are distinct:

Obligation	Supplied mechanism	Limit
Reject nondecreasing potential updates	Governor's `CUT` rule	Does not force a positive rational sequence to end

Obligation	Supplied mechanism	Limit
Bound attempted transitions	Demo 1's finite outer loop	The class itself has no fuel limit; `turn` is a counter
Bound a single external solver call	Native adapter's 30-second timeout	Does not time out the governor's in-process reference solver

A finite fuel budget guarantees at most that many attempts when every attempt returns or times out. Exhaustion with a positive residual means unfinished work. A forest-count cap is a count bound, not an elapsed-time bound. A fixed cost lattice with a positive spacing and a lower bound can bound admitted strict descents, but rejected attempts and hanging calls still need limits.

Problem 3.2

For a connected graph, rank $B = n-1$, so **some** optimal residual has support at most $m - n + 1 = \beta_1$. This is an existence statement; a formulation in residual variables is needed when referring to a basic feasible LP solution.

The equal-weight triangle has multiple minimizers even after fixing $s_a = 0$. For $(0, -2, -1)$, residual $(3, 0, 0)$ costs 6. For $(0, 1, -1)$, residual $(0, 3, 0)$ also costs 6. Their potential difference is not constant. The circulation $(2, 2, 2)$ is feasible and has objective 6, proving optimality and excluding a claimed minimum of 2.

For $S = \operatorname{supp}(r^*)$, the sufficient weighted null-space condition is

$$\sum_{e \in S} w_e |z_e| < \sum_{e \notin S} w_e |z_e| \quad \text{for every } 0 \neq z \in \operatorname{im} B.$$

Take $S = \{e_1\}$ and $z = B(0, 1, 0) = (1, -1, 0)$. Both sides equal 2, so the strict condition fails. The two exhibited optima prove nonuniqueness on this triangle. Failure of the sufficient condition **by itself** does not prove nonuniqueness for a particular residual; Challenge A gives a counterexample to that invalid converse.

Further check: the zero potential gives dense residual $(1, 1, 1)$, also optimal. The residual optimum set is $\{r \geq 0 : \sum r_i = 3\}$, a two-dimensional simplex. Cycle rank 1 does not bound every optimal support or the optimal-face dimension.

Problem 3.3

K_6 has $6^4 = 1296$ spanning trees. K_8 has $8^6 = 262144$.

Source inspection shows that the Rust solver can check 100,000 complete forests, then returns `GraphPhi1Error::EnumerationLimit` when the next complete forest would exceed the cap. It returns neither an approximate value nor a partial minimum. If the adapter's separate timeout is reached first, the observed

Python outcome is `NativeFunctionalTimeout`; the count alone does not guarantee when the Rust enumeration will finish.

A partial minimum is an upper bound over the examined candidates. Without exhaustion of the proved finite search set or an independent lower-bound witness, it cannot be labeled the global minimum. The response's count is a record of the solver's claimed work, not an independently checked proof of exhaustive execution.

Problem 3.4

According to the manuscript, the local statements concern pair cancellation under supplied sign and weight hypotheses; impossibility of assigning triangle vertex signs making all three edge products -1; and a one-value policy type whose theorem holds by construction.

They do not imply a polynomial-time general graph algorithm, correctness of the Python or Rust production runtime, or a physical fermion sign-problem solution. They also do not imply termination of arbitrary rational descent or that the triangle theorem has been generalized to all odd cycles.

The narrow claim is that the named local files, when built with their pinned toolchain and documented axioms, check those local statements. The distributed package contains the Lean files in `lean4/`, whose clean build and axiom dependencies (`propext`, `Quot.sound`) are attested in `lean4/axiom_report.txt` and verified in `docs/release_validation_2026_10_02_review_repairs.json`.

For a physical bridge, specify the Hamiltonian and representation, the finite or continuous configuration carrier, and the mapping into cochains. Prove the precise relation between signs, weights, and the claimed observable or obstruction. A claimed algorithm also needs implementation correspondence and a complexity proof with a stated input encoding and cost model. A toy theorem, prose analogy, or file hash does not supply these steps.

Final challenge hints and solutions

Challenge A

Hint: when $|\lambda_e| < w_e$, equality in weak duality forces the residual on that edge to be zero. Ask whether those edges connect all vertices.

In edge order ab, ac, ad, bc, bd, cd, the zero potential has residual $(1, -1, 0, 0, 0, 0)$ and cost 2. The proposed dual has zero divergence at every vertex, is within the unit capacities, and has objective $1(1) + (-1)(-1) = 2$. Therefore the zero potential is optimal.

On ad, bc, bd, cd, the dual has strict capacity slack. Any optimum must therefore have zero residual on all four edges. Their reports are zero, so $s_a = s_d$, $s_b = s_c$, $s_b = s_d$, and $s_c = s_d$. The subgraph is connected; every potential is constant, and fixing $s_a = 0$ makes the zero potential the unique optimum.

For $h = (1, 0, 0, 0)$, $Bh = (-1, -1, -1, 0, 0, 0)$. With $S = \{ab, ac\}$, the two weighted sums are 2 and 1; the sufficient condition fails. This does not contradict the theorem: the condition guarantees uniqueness for every residual on the support, while this particular signed residual is unique without satisfying that stronger condition.

Challenge B

Hint: reject an inconsistent primal value before asking whether it is globally minimal.

The given potential has residual $(3, 0, 0)$ and cost 6, so it disagrees with the claimed `phi1=2`. The feasible circulation $(2, 2, 2)$ has objective 6, which also excludes a global minimum of 2.

Require a JSON object with the documented status and result shape; integer or fraction strings for every rational; a gauge covering exactly the declared vertices; an integer forest count in the declared range, excluding Boolean values; nonnegative `phi1`; and equality between the reported value and the exactly recomputed weighted residual cost. Handle refusals, malformed replies, nonzero exits, and timeouts as typed failures, without substituting another solver's answer after a native failure.

Those consistency checks do not prove a lower bound or that enumeration really exhausted its search set. The caller still relies on the native implementation and its source-to-binary custody, or must check a feasible dual witness achieving the same value. The supplied native response has no dual. A hash can identify bytes; it does not prove this optimization theorem.

Paper 001 notebook learning worksheet

Teaching companion for the v0.1 preprint. Attempt each exercise before reading the answers below. All assessed arithmetic uses exact integers and fractions. The original Landau examination remains the main course; these exercises connect its ideas to the actual notebooks and Lean source.

Landau puts the three notebooks on the table. "A printed PASS is the beginning of your inspection, not the end. Tell me what was checked." This dialogue is fictional.

Six exercises

1 Certify the second triangle

Demo 1 uses reports `(1,2,2)` and penalties `(1,2,2)`, in edge order `ab,bc,ca`. Compute the zero-potential cost. Find a potential with minimum cost, give its residual, and supply a balanced capacity-bounded dual with the same objective. Explain why its answer differs from the manuscript's `3`.

Hint: keep the last two reports exact and put the residual on the cheapest edge.

2 Protect the fractional potential

Change the reports to `(1,1/2,2)` while retaining penalties `(1,2,2)`. Construct a matching primal/dual pair. Convert your potential values with `int(v)` and recompute the cost. Explain what information was lost and replace the notebook's proposal serialization with an exact alternative.

Hint: the two non-cheapest edges force `s_c=-2` and `s_b=-5/2` when `s_a=0`.

3 Four laws are not coverage

The host declares weights `n1=-1/2`, `p1=1/2`, `n2=-3/4`, and `residue=2`. A proposer pairs `n1` with `p1` and leaves the other IDs unpaired.

Check involution, absence of fixed points, sign reversal, and equal magnitude on that paired subset. Should the gate admit this as a cancellation certificate for the whole host carrier? Repair the host example by adding one positive weight and extending the pairing. What weight remains after cancellation?

Hint: inspect `negative_coverage` separately from the four pair laws.

4 Read the Lean hypotheses

In `InvolutionLaws.lean`, identify the four fields of `InvolutionWitness`. Which two are actually used by `pair_cancellation`? Does this theorem exhibit a physical involution or establish a global finite-carrier sum? Is the type `SignedConfig` finite?

Challenge: explain why a two-term cancellation calculation is weaker than a proof that a finite carrier is partitioned into valid pairs.

5 Inspect the policy theorem

`AuthorityEffect` has a single constructor, `none`. The theorem accepts a packet and a premise that it was admitted. Does the proof use that premise? Could the packet type represent an admitted packet with a different authority effect? State a stronger proposition that follows from the type definition, and name the separate runtime obligation.

Hint: look at `cases p.authority`, then inspect the constructors.

6 Audit the debate receipt

The ten stored trials have TP=6, FP=2, TN=2, and FN=0 against historical verifier labels. Compute precision, recall, and accuracy as fractions. On fresh replay, the current gate refuses all ten proposals. Does either result establish physical ground truth or show the proposer false?

The recovered source trace has CRLF line endings. Its raw hash differs from the notebook's recorded hash, but converting only CRLF to LF in an isolated copy exactly recovers that recorded hash. Explain what this establishes, what should remain unchanged, and which bytes a release receipt must identify.

Worked answers

1 Second triangle

The zero-potential residual is `(1,2,2)`, so the cost is `1+4+4=9`. Potential `(0,-4,-2)` has fitted differences `(-4,2,2)` and residual `(5,0,0)`, costing `5`. The dual `(1,1,1)` is balanced, fits capacities `(1,2,2)`, and has objective `1+2+2=5`. Equal bounds give the minimum `5`. The manuscript uses different reports and weights; it is a separate minimum-3 instance.

2 Fractional potential

Potential `(0,-5/2,-2)` has differences `(-5/2,1/2,2)` and residual `(7/2,0,0)`, costing `7/2`. The feasible unit circulation has objective `1+1/2+2=7/2`.

Integer conversion truncates `-5/2` toward zero, giving `(0,-2,-2)`. Its residual is `(3,1/2,0)`, costing `3+2(1/2)=4`. The conversion loses a half-unit in the potential.

Use integer/fraction strings at the proposal boundary:

```
good_cut = gov.execute_move(
    MoveType.CUT,
    {"potentials": {k: str(v) for k, v in best_pot.items()}},
)
```

This preserves exact values, including after rational edits. For a complete lesson, obtain the primal/dual certificate and verify its coverage, balance, capacities, and matching objectives. A computed minimum and a zero obstruction remain different claims.

3 Coverage

The supplied pair satisfies all four laws on `{n1,p1}`. It fails negative coverage because `n2` is an unpaired negative weight. The current gate refuses it. Add `p2=3/4` and pair `n2` with `p2` in both directions. Both pairs cancel, leaving the nonnegative residue `2`. Do not invent an undeclared weight in a real host carrier: this repair is a new host-owned toy example, not authorization for a proposer to alter the original instance.

4 Conditional pair theorem

The fields are `involution_law`, `fixed_point_free`, `sign_reversing`, and `weight_preserving`. The arithmetic proof uses the latter two. It assumes a supplied witness; it does not construct one for a physical representation.

`SignedConfig` has natural-number IDs and weights and therefore is not a finite type. The theorem adds two terms, not a total carrier sum. A global pair-partition result needs a declared carrier, closure and involution evidence, and coverage of the configurations whose signs are being canceled. The Python finite gate additionally checks that every negative host weight is covered and that the unpaired residue is nonnegative.

5 By construction

The admission premise `_h` is unused. Every packet representable by this type has authority `none`, regardless of its admission Boolean. The stronger proposition is `for every p : PacketEnvelope, p.authority = AuthorityEffect.none`. The single-constructor definition gives this equality; it cannot establish that a separate runtime never performs an authorized external action. That requires a correspondence between the implementation, its transitions, and the stated policy.

6 Labels and custody

Precision is `6/(6+2)=3/4`, recall is `6/(6+0)=1`, and accuracy is `(6+2)/10=4/5`. These quantities describe agreement with historical labels. Ten present refusals establish that the proposals do not meet the current checks; refusal is not a theorem that the proposed physical conclusion is false. Neither set of labels is independent physical ground truth.

The line-ending conversion recovers the exact byte stream named by the stored hash. Preserve the original source and describe the recovered copy. A receipt must hash the actual bytes it claims to identify; mathematical equivalence of

parsed records does not make two byte streams identical. No live model call is needed to replay this deposited trace.